

JavaScript

Project Theme: Local Community Event Portal

#index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Local Community Event Portal</title>

  <!-- Google Font -->

  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap"

    rel="stylesheet">

  <!-- External CSS -->

  <link rel="stylesheet" href="styles.css">

</head>

<body>

  <h1 id="mainHeader">

    Local Community Event Portal

  </h1>

  <!-- Navigation -->

  <nav>

    <ul class="navMenu">

      <li><a href="#">Home</a></li>

      <li><a href="#">Events</a></li>

      <li><a href="#">Register</a></li>

      <li><a href="#">Contact</a></li>

    </ul>

  </nav>
```

```
<!-- Search -->

<section>

  <input

    type="text"

    id="searchBox"

    placeholder="Search Event">

</section>

<br>

<!-- Category Filter -->

<section>

  <select id="categoryFilter">

    <option value="All">All Categories</option>

    <option value="Music">Music</option>

    <option value="Sports">Sports</option>

    <option value="Workshop">Workshop</option>

  </select>

</section>

<br>

<!-- Event Cards -->

<section>

  <div id="eventContainer">

  </div>

</section>

<br>

<!-- Registration Form -->

<section>

  <h2>Register</h2>

  <form id="registrationForm">

    <input

      type="text"
```

```
        name="name"

        placeholder="Enter Name">

<br><br>

<input

        type="email"

        name="email"

        placeholder="Enter Email">

<br><br>

<select name="event">

        <option>Music Festival</option>

        <option>Sports Meet</option>

        <option>Baking Workshop</option>

</select>

<br><br>

<button

        type="submit"

        id="registerBtn">

        Register

</button>

</form>

<p id="message"></p>

</section>

<!-- Loading Message -->

<div id="loading">

        Loading Events...

</div>

<!-- jQuery Library -->

<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

<!-- JavaScript File -->

<script src="main.js"></script>
```

```
</body>
```

```
</html>
```

#styles.css

```
/* =====  
  
LOCAL COMMUNITY EVENT PORTAL  
  
Complete Styles.css  
  
===== */  
  
/* Reset */  
  
* {  
    margin:0;  
    padding:0;  
    box-sizing:border-box;  
}  
  
/* Body */  
  
body {  
    font-family:'Poppins',sans-serif;  
    background:#e8f5e9 url("images/background.jpg");  
    background-size:cover;  
    background-repeat:no-repeat;  
    background-attachment:fixed;  
    color:#333;  
    line-height:1.6;  
    padding:20px;  
}  
  
/* Header */  
  
#mainHeader {  
    background:linear-gradient(to right,#007BFF,#0056b3);
```

```
color:white;

text-align:center;

padding:25px;

font-size:42px;

font-weight:bold;

border-radius:10px;

margin-bottom:25px;

}

/* Navigation */

nav{

    background:white;

    padding:15px;

    border-radius:8px;

    margin-bottom:25px;

    box-shadow:0 2px 6px rgba(0,0,0,0.2);

}

.navMenu{

    display:flex;

    justify-content:center;

    list-style:none;

    gap:30px;

}

.navMenu li{

    list-style:none;

}

.navMenu a{

    text-decoration:none;

    color:#007BFF;

    font-weight:bold;

    font-size:18px;
```

```
}

.navMenu a:hover{

    color:red;

}

/* Sections */

section{

    margin-bottom:25px;

}

/* Search Box */

#searchBox{

    width:320px;

    padding:10px;

    font-size:16px;

    border:1px solid gray;

    border-radius:6px;

}

/* Category */

#categoryFilter{

    width:220px;

    padding:10px;

    font-size:16px;

    border-radius:6px;

}

/* Event Container */

#eventContainer{

    display:flex;

    flex-wrap:wrap;

    gap:20px;

}

/* Event Card */
```

```
.eventCard{

    width:300px;

    background:white;

    border-radius:10px;

    padding:20px;

    box-shadow:0 2px 8px rgba(0,0,0,0.2);

    transition:0.3s;

}

.eventCard:hover{

    transform:scale(1.03);

}

.eventCard h3{

    color:#007BFF;

    margin-bottom:10px;

}

.eventCard p{

    margin:8px 0;

}

/* Buttons */

button{

    background:#007BFF;

    color:white;

    border:none;

    padding:10px 20px;

    border-radius:5px;

    cursor:pointer;

    margin-top:10px;

    font-size:15px;

}

button:hover{
```

```
background:#0056b3;
}
/* Register Heading */
h2{
margin-bottom:15px;
color:#007BFF;
}
/* Form */
form{
background:white;
padding:20px;
border-radius:10px;
width:400px;
box-shadow:0 2px 6px rgba(0,0,0,0.2);
}
input{
width:100%;
padding:10px;
border:1px solid gray;
border-radius:5px;
font-size:15px;
}
input:focus{
outline:3px solid orange;
}
select{
width:100%;
padding:10px;
border-radius:5px;
}
```



```
/* Success Message */

#message{

    margin-top:15px;

    font-weight:bold;

}

/* Loading */

#loading{

    display:none;

    font-size:20px;

    color:red;

    margin-top:20px;

}

/* Table */

table{

    width:100%;

    border-collapse:collapse;

    margin-top:20px;

    background:white;

}

th{

    background:#007BFF;

    color:white;

    padding:12px;

    border:1px solid gray;

}

td{

    padding:12px;

    border:1px solid gray;

    text-align:center;

}
```

```
tr:nth-child(even){
    background:#f2f2f2;
}

/* Multi-column News */
.news{
    column-count:2;
    column-gap:30px;
    column-rule:1px solid gray;
    background:white;
    padding:20px;
    border-radius:8px;
}

/* Images */
img{
    max-width:100%;
    border-radius:8px;
}

/* Visibility Examples */
.hidden{
    visibility:hidden;
}

.none{
    display:none;
}

/* Responsive */
@media screen and (max-width:768px){
    #mainHeader{
        font-size:28px;
    }
}

.navMenu{
```

```
    flex-direction:column;

    align-items:center;
  }

  #eventContainer{

    flex-direction:column;

  }

  .eventCard{

    width:100%;

  }

  form{

    width:100%;

  }

  .news{

    column-count:1;

  }

  #searchBox{

    width:100%;

  }

  #categoryFilter{

    width:100%;

  }
}
```

#main.js

```
// =====

// Exercise 1 : JavaScript Basics

// =====

console.log("Welcome to the Community Portal");
```

```
window.onload = function ()

{

    alert("Page Loaded Successfully");

};

// =====

// Exercise 2 : Variables, Data Types and Operators

// =====

const eventName = "Music Festival";

const eventDate = "20 August 2026";

let availableSeats = 5;

console.log(`Event : ${eventName}`);

console.log(`Date : ${eventDate}`);

console.log(`Seats : ${availableSeats}`);

// Seat Count

availableSeats--;

console.log(`Seats After Registration : ${availableSeats}`);

// =====

// Exercise 5 : Objects and Prototypes

// =====

class Event

{

    constructor(id, name, category, location, seats)

    {

        this.id = id;

        this.name = name;

        this.category = category;

        this.location = location;

        this.seats = seats;

    }

}
```

```

// Prototype Method

Event.prototype.checkAvailability = function ()
{
    return this.seats > 0;
};

// =====

// Events Array

// =====

let events =
[
    new Event(1, "Music Festival", "Music", "City Park", 5),
    new Event(2, "Sports Meet", "Sports", "Sports Complex", 0),
    new Event(3, "Baking Workshop", "Workshop", "Community Hall", 10),
    new Event(4, "Dance Show", "Music", "Open Ground", 15)
];

// Display Object Entries

console.log("Object Entries");

Object.entries(events[0]).forEach(entry =>
{
    console.log(entry[0], entry[1]);
});

// =====

// Exercise 3 : Conditionals, Loops, Error Handling

// =====

function displayEvents(eventList)
{
    const container = document.querySelector("#eventContainer");

    container.innerHTML = "";

    eventList.forEach(event =>
    {

```

```

    if (event.seats > 0)
    {
        const card = document.createElement("div");

        card.className = "eventCard";

        card.innerHTML =
        `
        <h3>${event.name}</h3>

        <p>Category : ${event.category}</p>

        <p>Location : ${event.location}</p>

        <p>Seats : ${event.seats}</p>

        <button onclick="registerUser(${event.id})">

        Register

        </button>

        `;

        container.appendChild(card);
    }
});
}

// =====

// Exercise 4 : Functions

// =====

// Add Event

function addEvent(id, name, category, location, seats)
{
    events.push(
        new Event(id, name, category, location, seats)
    );
}

// Register User

function registerUser(id)

```

```

{
  try
  {
    let event = events.find(e => e.id === id);

    if (!event)
    {
      throw "Event Not Found";
    }

    if (event.seats <= 0)
    {
      throw "No Seats Available";
    }

    event.seats--;

    alert("Registration Successful");

    displayEvents(events);
  }
  catch(error)
  {
    alert(error);
  }
}

// Filter Events

function filterEventsByCategory(category, callback)
{
  let filtered;

  if(category === "All")
  {
    filtered = events;
  }

  else

```

```

{
    filtered = events.filter(
        event => event.category === category
    );
}

callback(filtered);
}

// =====

// Closure Example

// =====

function registrationCounter()
{
    let total = 0;

    return function()
    {
        total++;

        console.log("Total Registrations : " + total);
    }
}

const countRegistration = registrationCounter();

// =====

// Initial Display

// =====

displayEvents(events);

// =====

// Exercise 6 : Arrays and Methods

// =====

// Add a new event using push()

addEvent(5, "Coding Workshop", "Workshop", "Library Hall", 20);

console.log("All Events");

```



```
console.log(events);

// Filter only Music Events

const musicEvents = events.filter(event => event.category === "Music");

console.log("Music Events");

console.log(musicEvents);

// Map Event Names

const eventCards = events.map(event => `Workshop on ${event.name}`);

console.log("Formatted Event Names");

console.log(eventCards);

// =====

// Exercise 7 : DOM Manipulation

// =====

// Refresh UI

function refreshUI()

{

    displayEvents(events);

}

refreshUI();

// =====

// Exercise 8 : Event Handling

// =====

// Category Filter

document.querySelector("#categoryFilter")

.addEventListner("change", function ()

{

    filterEventsByCategory(this.value, displayEvents);

});

// Search Events

document.querySelector("#searchBox")

.addEventListner("keydown", function ()
```

```

{

let keyword = this.value.toLowerCase();

let filtered = events.filter(event =>

    event.name.toLowerCase().includes(keyword)

);

displayEvents(filtered);

});

// =====

// Exercise 9 : Promises

// =====

function loadEventsPromise()

{

    document.querySelector("#loading").style.display = "block";

    return new Promise((resolve) =>

    {

        setTimeout(() =>

        {

            resolve(events);

        }, 2000);

    });

}

loadEventsPromise()

.then(result =>

{

    console.log("Promise Loaded");

    displayEvents(result);

})

.catch(error =>

{

    console.log(error);

```

```

})

    .finally(() =>
    {
        document.querySelector("#loading").style.display = "none";
    });

// =====
// Exercise 9 : Async Await
// =====

async function loadEventsAsync()
{
    document.querySelector("#loading").style.display = "block";

    try
    {
        let result = await loadEventsPromise();

        console.log("Async Await Loaded");

        displayEvents(result);
    }

    catch(error)
    {
        console.log(error);
    }

    finally
    {
        document.querySelector("#loading").style.display = "none";
    }
}

loadEventsAsync();

// =====
// Exercise 10 : Modern JavaScript
// =====

```

```
// Destructuring

const {
  name,
  category,
  location: eventLocation
} = events[0];

console.log(name);
console.log(category);
console.log(eventLocation);

// Spread Operator

const clonedEvents = [...events];

console.log("Cloned Events");
console.log(clonedEvents);

// Default Parameters

function showEvent(eventName = "Community Event")
{
  console.log(eventName);
}

showEvent();
showEvent("Music Festival");

// =====

// Exercise 11 : Working with Forms

// =====

const form = document.querySelector("#registrationForm");

form.addEventListener("submit", function (event)
{
  event.preventDefault();

  const name = form.elements["name"].value.trim();

  const email = form.elements["email"].value.trim();

  const selectedEvent = form.elements["event"].value;
```

```

const message = document.querySelector("#message");

if(name === "" || email === "")
{
    message.style.color = "red";

    message.innerHTML = "Please fill all fields.";

    return;
}

message.style.color = "green";

message.innerHTML =

`Registration Successful for ${selectedEvent}`;
});

// =====

// Exercise 12 : AJAX & Fetch API

// =====

// GET Request

fetch("events.json")

.then(response => response.json())

.then(data =>

{

    console.log("Events From JSON");

    console.log(data);

})

.catch(error =>

{

    console.log(error);

});

// POST Request

function sendRegistration()

{

    const user =

```

```
{
  name:"John",
  email:"john@gmail.com"
};

document.querySelector("#loading").style.display="block";

setTimeout(()=>
{
  fetch("https://jsonplaceholder.typicode.com/posts",
  {
    method:"POST",
    headers:
    {
      "Content-Type":"application/json"
    },
    body:JSON.stringify(user)
  })
  .then(response=>response.json())
  .then(data=>
  {
    alert("Registration Sent Successfully");
    console.log(data);
  })
  .catch(error=>
  {
    alert("Submission Failed");
    console.log(error);
  })
  .finally(()=>
  {
    document.querySelector("#loading").style.display="none";
```

```

    });

    },2000);
}

sendRegistration();

// =====

// Exercise 13 : Debugging

// =====

console.log("Debugging Started");

console.log("Events Array");

console.log(events);

console.log("Registration Form");

console.log(form);

// =====

// Exercise 14 : jQuery

// =====

$("#registerBtn").click(function()
{
    $(".eventCard").fadeOut(1000);

    $(".eventCard").fadeIn(1000);
});

console.log("Benefit of React/Vue");

console.log("Reusable Components and Better State Management");

```

#events.json

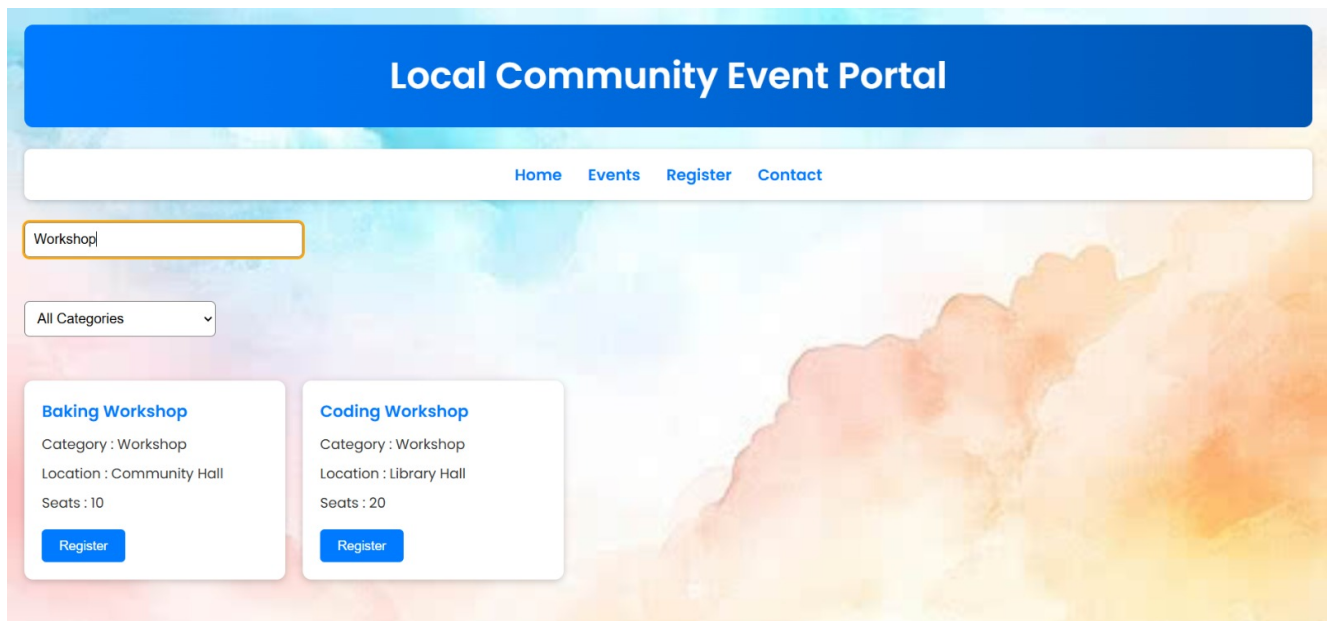
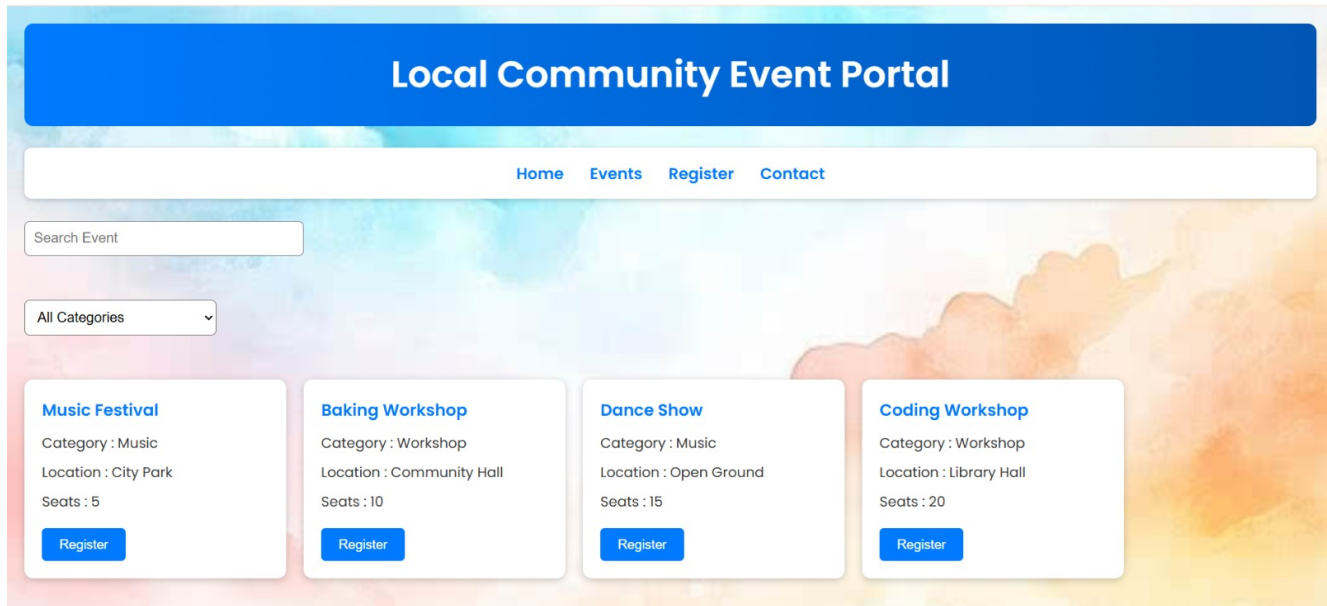
```

[
  {
    "id":1,
    "name":"Music Festival",

```

```
    "category": "Music",  
    "location": "City Park",  
    "seats": 10  
  },  
  {  
    "id": 2,  
    "name": "Sports Meet",  
    "category": "Sports",  
    "location": "Sports Ground",  
    "seats": 15  
  },  
  {  
    "id": 3,  
    "name": "Baking Workshop",  
    "category": "Workshop",  
    "location": "Community Hall",  
    "seats": 20  
  }  
]
```


Output:



Register



Registration Successful for Music Festival

Local

127.0.0.1:5500 says
Registration Successful

OK

ortal

[Home](#) [Events](#) [Register](#) [Contact](#)

Music Festival

Category : Music
Location : City Park
Seats : 5

Baking Workshop

Category : Workshop
Location : Community Hall
Seats : 10

Dance Show

Category : Music
Location : Open Ground
Seats : 15

Coding Workshop

Category : Workshop
Location : Library Hall
Seats : 20